

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460324.7252 +/- 0.0056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.016
Transit depth (Rp/Rs)^2	2.11 +/- 0.46 [%]
Orbital Inclination (inc)	86.61 +/- 1.03
Airmass coefficient 1 (a1)	1.0305 +/- 0.0093
Airmass coefficient 2 (a2)	-0.017 +/- 0.039
Scatter in the residuals of the lightcurve fit is	0.9 %
Best Comparison Star	#2 - [181, 337]
Optimal Aperture	3.74
Optimal Annulus	10.86
Transit Duration (day)	0.09 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.145 ± 0.016 vs 0.14195069566578392 (deviation 0.19σ)
Duration fit vs published	0.09 d vs 0.0875 d (deviation 0.1σ)
Mid-time O–C	-20.0 min
Depth SNR	9.1
Residual scatter	0.9 %

Light curve

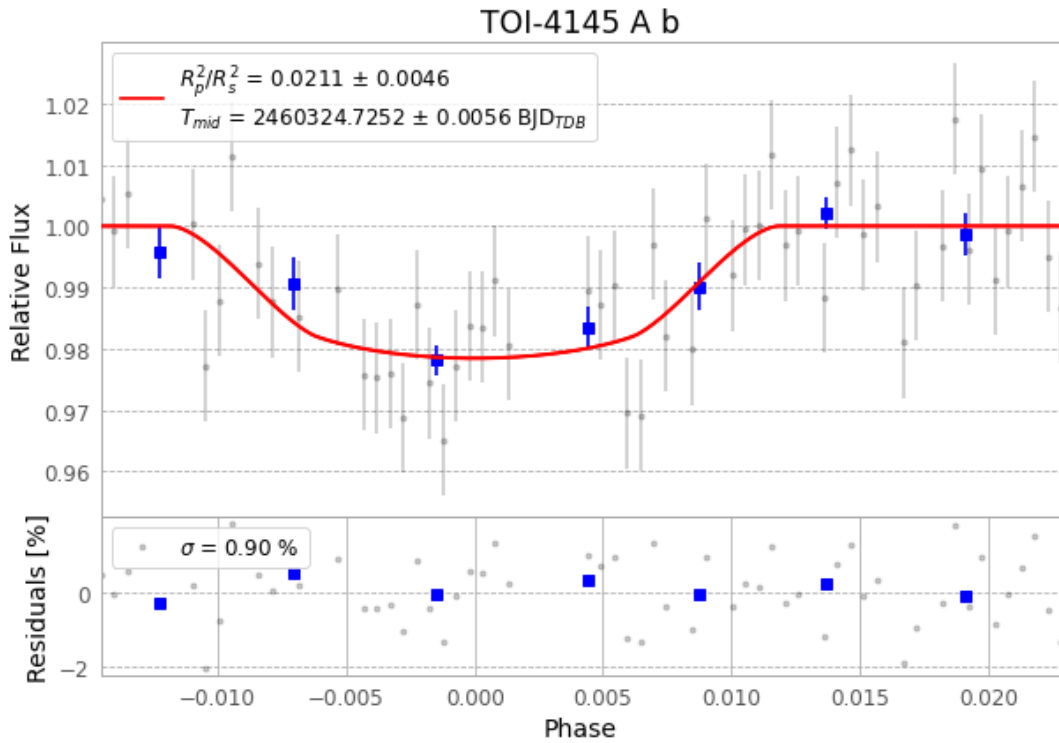


Figure 1 — FinalLightCurve_TOI-4145 A b_2024-01-14.png (SHA-256 9ec174765457dec9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [309, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6199fa1fd6796296...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

74 FITS frames (49 MB), TOI4145240115035415.FITS → TOI4145240115073315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4145-240115.log (08724d4de1d7...), FinalLightCurve_TOI-4145 A b_2024-01-14.png (9ec174765457...), FinalLightCurve_TOI-4145 A b_2024-01-14.csv (1337d5fa9e09...)

Compute resources

Wall time 146.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 00:31Z.